

SHORYA GUPTA

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EDUCATION

Thapar Institute of Engineering and Technology

Aug 2023 – Jul 2027

Bachelor of Engineering, Computer Science Engineering

CGPA: 8.87 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Parallel Computing

ACHIEVEMENTS

- **Codeforces Expert** (Max Rating: 1687) across 16 rated contests
- **LeetCode Knight** (Peak Contest Rating: 2105) across 26 rated contests
- **SPHINX'24 Hackathon Runner-Up** among 2500+ individual registrations, hosted by MNIT Jaipur
- **Runner-Up, Inter-College Coding Contest**, Thapar Institute
- **Merit Scholarship, Thapar Institute** — GPA-based award for academic excellence (9.34 AGPA)

SKILLS

Languages: C++, Python, SQL

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, Async Python

Data Stores: PostgreSQL, Redis, pgvector

AI Engineering: LangGraph, RAG, Vector Search, Embeddings, LLM APIs

Distributed Systems: Apache Kafka, Event-Driven Architecture, Concurrency Control, Idempotency, Rate Limiting

DevOps: Docker, Docker Compose, GitHub Actions, CI, Prometheus, Grafana, Linux

TECHNICAL PROJECTS

DeliverIQ | Distributed Order Dispatch API [GitHub]

2026

Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Kafka, Docker, Prometheus, Grafana

- Eliminated a double-dispatch race across 3 API replicas with a two-phase **SELECT FOR UPDATE SKIP LOCKED** claim protocol; verified zero duplicate assignments under concurrent load
- Built the dispatch core: aging-weighted priority scheduler preventing starvation, and geohash matching with a fairness band that spreads work across idle riders
- Streamed events to 3 Kafka consumer groups with manual offset commits for at-least-once delivery, idempotent consumption on (partition, offset), and a dead-letter queue for poison messages
- Hardened after a security audit: closed unauthenticated endpoints, a forgeable admin JWT, and a cross-tenant idempotency leak; added RBAC, Lua-atomic rate limiting, and 71 integration tests

DocMind | Agentic RAG Document Assistant [GitHub]

2026

Python, LangGraph, Groq, Sentence-Transformers, pgvector, PostgreSQL, Redis, FastAPI

- Built a document Q&A pipeline: chunked PDF ingestion, Sentence-Transformers embeddings, and pgvector similarity search returning answers with cited source chunks
- Built a self-correcting retrieval loop in LangGraph (retrieve, grade the chunks, rewrite the query, answer) so a poor first search retries instead of answering from irrelevant context
- Added a semantic query cache (Redis vector search) serving paraphrased repeat questions without an LLM call, with a tuned similarity threshold

EXTRACURRICULAR ACTIVITIES

- Vice Head, Photography, FAPS Society (Fine Arts & Photography Society), Thapar Institute
- State-level football player (2022); long jump athlete, college athletics team at inter-college competitions